

A Reconfigurable Energy Storage Architecture for Energy-harvesting Devices

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Introduction

- Run entirely on energy collected from the environment (solar, RF, vibration) without a battery
- Enables maintenance-free deployment in extreme or remote environments (glaciers, orbit, etc)
- Operate intermittently by accumulating charge, running until depleted, powering off, recharging, and repeating

Problem

- Buffer energy in a capacitor and run only while charge lasts
- Fixed-size buffer cannot satisfy conflicting task needs
- Small buffer provides fast recharge and good reactivity but too little energy for demanding atomic tasks
- Large buffer provides enough energy but long recharge leaves device off and unresponsive
- Tasks needing both instant reaction and high energy are unsupported

Solution

- Co-designed hardware and software power system with runtime-reconfigurable energy storage
- Software declares per-task energy needs while hardware reconfigures capacity to match

Key Innovation

1. Software:

- `config(mode)` charges to a specific capacity before running a capacity-constrained task
- `burst(mode)` runs instantly using energy pre-charged earlier for reactive and temporal tasks
- `preburst(bmode, emode)` pre-charges a future burst task's bank ahead of time and off the critical path

2. Hardware:

- Array of capacitor banks connected via state-retaining switches
- Software controls total capacitance rather than voltage thresholds
- Faster to cold-start and cheaper in area and leakage

Results

- Achieves 2 to 4 times better event-detection accuracy vs fixed-capacity systems
- Deployed on a nano-satellite (CapySat) under extreme size and temperature limits using a simplified switch at 20% the area

Conclusion

- Fixed buffers force a tradeoff between atomicity and reactivity
- Cappybara removes this by reconfiguring capacity and pre-charging bursts at runtime
- Enables reactive applications previously impossible on energy-harvesting hardware